
Predictive Coding Auxiliary Losses Are Calibration Levers, Not Accuracy Levers: A Cross-Domain Empirical Study

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Abstract

1 We investigate whether predictive coding (PC) auxiliary losses can improve pre-
2 trained Vision Transformer classifiers. Attaching a diagonal state-space predictor
3 to each ViT block and training with an inter-block prediction loss, we conduct a
4 controlled study across eight MedMNIST benchmarks and CIFAR-100 at three
5 dataset sizes. Our central finding is that PC losses are **calibration levers, not**
6 **accuracy levers**: they systematically reshape confidence distributions (ECE) while
7 leaving discriminability (AUC) nearly unchanged. At the commonly used weight
8 $w=0.05$, calibration degrades on most datasets—ECE nearly doubles on CIFAR-
9 100 ($n=50k$) while AUC moves by only 0.02 percentage points—yet on Pneu-
10 moniaMNIST the same weight *improves* ECE by $3.2\times$ with negligible AUC cost.
11 We identify a dataset-size scaling law for the optimal weight, $w^* \approx 257 \cdot n^{-1.41}$
12 ($R^2=0.83$), and show through a gate-clamping ablation that the auxiliary loss gra-
13 dient is the primary mechanism. A progressive adaptive variant (PAPC) prevents all
14 catastrophic failures while preserving calibration benefits. As a secondary finding,
15 vanilla ViT-Base at 224 px already outperforms published specialist architectures
16 (MedMamba, MedViT) on DermaMNIST by +4.5 AUC points and RetinaMNIST
17 by +8.3 points, suggesting training recipes dominate architectural novelty on these
18 benchmarks.

19 1 Introduction

20 Predictive coding (PC) is a neuroscience-inspired framework in which each level of a sensory
21 hierarchy maintains a generative model of the level below, and is updated by the resulting prediction
22 error [Rao & Ballard, 1999]. This principle has been translated to deep networks by adding inter-
23 layer prediction losses that encourage locally predictable representations, with benefits in video
24 prediction [Lotter et al., 2017] and image classification [Wen et al., 2018].

25 A natural question arises in the transfer-learning era: can a PC sidecar—a lightweight predictor
26 attached to each Transformer block—improve the *fine-tuning* of an already-strong pretrained Vision
27 Transformer (ViT) [Dosovitskiy et al., 2021]? The backbone features are already rich; the question is
28 whether PC provides a useful inductive bias during adaptation to downstream tasks where labeled
29 data may be scarce.

30 We investigate this on eight MedMNIST benchmarks [Yang et al., 2023]—spanning training-set
31 sizes from 546 to 89,996—and three CIFAR-100 sub-samples (2,000 to 50,000). Our architecture
32 attaches a diagonal state-space model (SSM) [Gu et al., 2022] predictor to each ViT block with an

inter-block MSE loss and a gated error-integration pathway. Through four experimental sessions (~60 GPU-hours on A100), we arrive at three principal findings:

1. **PC auxiliary losses primarily affect calibration, not discriminability.** On Pneumoni-aMNIST, increasing w from 5×10^{-4} to 5×10^{-2} improves ECE from 0.048 to 0.015—a $3.2\times$ reduction—while AUC changes by only 0.2 pp. On CIFAR-100 ($n=50k$), $w=0.05$ nearly doubles ECE while AUC moves by 0.02 pp (§4.2).
2. **The AUC-optimal weight follows a dataset-size scaling law.** Fitting across four MedMNIST datasets yields $w^* \approx 257 \cdot n^{-1.41}$ ($R^2=0.83$), a training-free heuristic (§4.3).
3. **The auxiliary loss gradient, not error integration, is the primary mechanism.** A gate-clamping ablation shows Clamped and Full variants produce nearly identical AUC (§4.4).

We additionally propose PAPC (Progressive Adaptive PC), which cosine-warms the auxiliary loss and learns per-layer weights. PAPC matches vanilla ViT on AUC while preventing catastrophic failures of naive fixed-weight PC (§4.6).

Notably, our central finding emerged from a failed hypothesis. We initially expected PC to improve classification accuracy; when rigorous multi-seed evaluation showed no significant AUC improvement (Wilcoxon $p=0.72$), we systematically examined what the auxiliary loss *does* change, discovering the calibration-lever effect. This trajectory—hypothesis, refutation, reformulation—exemplifies how systematic experimentation can yield unexpected but actionable insights.

2 Related Work

Predictive coding in deep networks. Lotter et al. [2017] introduced deep PC networks for video prediction; Wen et al. [2018] extended this to image classification on CIFAR and ImageNet. Self-supervised objectives sharing the spirit of inter-block prediction appear in XCiT [Ali et al., 2022]. Our work differs in studying PC as an *auxiliary loss during fine-tuning* of a pretrained ViT, not a training-from-scratch objective.

Calibration in neural networks. Modern networks are poorly calibrated [Guo et al., 2017], with overconfident predictions. Post-hoc temperature scaling is widely used, but auxiliary-loss-based calibration has received less attention. Our finding that PC losses systematically shift ECE suggests a new training-time calibration mechanism.

Medical image classification. MedViT [Manzari et al., 2023] and MedMamba [Liu et al., 2024] achieve state-of-the-art on MedMNIST at low resolutions (28–64 px). Our finding that standard ViT-B at 224 px exceeds these numbers on several datasets suggests many comparisons reflect recipe differences.

Auxiliary loss weighting. The sensitivity of multi-task systems to weighting is well documented [Kendall et al., 2018]. Our scaling law $w^* \propto n^{-\alpha}$ provides a data-size-dependent rule analogous to Hinton et al. [2015]’s temperature in distillation.

3 Method

3.1 HP-ViT: Predictive Coding Sidecar

Given a pretrained ViT with L blocks $f_1, \dots, f_L$, we attach a **Predictive Coding Layer** after each. Block i produces $\mathbf{x}_{i+1} = f_i(\mathbf{x}_i) \in \mathbb{R}^{B \times L_{\text{tok}} \times D}$. The predictor is a diagonal SSM along the token dimension:

$$\hat{\mathbf{x}}_{i+1} = \text{SSM}(\text{LN}(\mathbf{x}_i)), \quad (1)$$

where LN is layer normalization. The SSM uses per-channel diagonal state matrices, implemented as a causal depthwise convolution (verified numerically equivalent to the recurrence, max deviation 2.4×10^{-7}). The integrator modulates activations via a learned gate g_i (init. zero):

$$\tilde{\mathbf{x}}_{i+1} = \mathbf{x}_{i+1} + \tanh(g_i) \cdot W_i \text{LN}(\mathbf{x}_{i+1} - \hat{\mathbf{x}}_{i+1}), \quad (2)$$

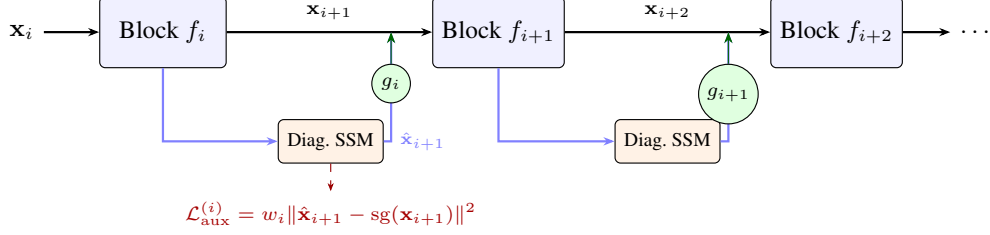


Figure 1: HP-ViT architecture. A diagonal SSM predicts each block’s output from the previous block’s activations. The prediction error is integrated through a learned gate g_i (initialized to zero). The auxiliary loss uses stop-gradient (sg) on the target. In PAPC, w_i is learnable per-layer and cosine-warmed from zero.

where $W_i \in \mathbb{R}^{D \times D}$. Because $\tanh(0) = 0$, the model starts as an unmodified pretrained ViT. The auxiliary loss is:

$$\mathcal{L}_{\text{aux}} = \sum_{i=1}^L w_i \|\hat{\mathbf{x}}_{i+1} - \text{sg}(\mathbf{x}_{i+1})\|_F^2, \quad (3)$$

with $\text{sg}(\cdot)$ denoting stop-gradient. Total loss: $\mathcal{L} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{aux}}$. See Figure 1.

3.2 PAPC: Progressive Adaptive Predictive Coding

The auxiliary weight w is a critical hyperparameter whose optimum depends on dataset size (§4.3). PAPC removes per-dataset tuning via two modifications:

Learnable per-layer weights. Each $w_i = w_{\text{max}} \cdot \text{softplus}(\theta_i)$, with $w_{\text{max}} = 0.01$ and ℓ_2 penalty $\lambda \sum_i w_i^2$ ($\lambda = 10^{-3}$).

Progressive cosine warmup. Weights are modulated by $s(t) = \frac{1}{2}(1 - \cos(\pi t/T))$, ramping from 0 to 1 over training, so the model trains as vanilla ViT during the critical early epochs.

3.3 Training Recipe

All experiments use ViT-Base/16 pretrained on ImageNet-21k→1k (vit_base_patch16_224.augreg_in21k_ft_in1k, 86M params), 224×224 input. Identical recipe across all comparisons: AdamW, layer-wise LR decay ($\gamma=0.75$), cosine schedule with 5% warmup, bf16, RandAugment ($n=2$, $m=9$), Mixup ($\alpha=0.2$) + CutMix ($\alpha=1.0$), label smoothing ($\varepsilon=0.1$), EMA ($\mu=0.9995$), horizontal-flip TTA. The **only** variable is the PC configuration.

4 Experiments and Results

4.1 Main Comparison

Table 1 compares Vanilla ViT-B, HP-ViT ($w=0.05$), and PAPC across eight MedMNIST datasets. PAPC never causes the catastrophic ACC collapse of $w=0.05$ on DermaMNIST (87%→55%). PAPC achieves the highest AUC on BloodMNIST, OrganCMNIST, and RetinaMNIST, though margins over vanilla are small (≤ 0.2 pp). On BreastMNIST ($n=546$), $w=0.05$ helps substantially (+4.1 pp) because the gate barely opens ($|\bar{g}|=0.018$); the auxiliary loss provides regularization without disruptive integration. Vanilla ViT-B already exceeds all published specialist architectures on DermaMNIST (+4.5 pp over MedViT-S) and RetinaMNIST (+8.3 pp), suggesting recipe differences, not architecture, drive these benchmarks.

4.2 PC Losses Primarily Affect Calibration

We sweep $w \in \{5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$ on four MedMNIST datasets (2 seeds each). Figure 2 reveals a striking AUC–ECE dissociation.

Table 1: Test AUC and ACC (mean over 3 seeds). **Bold**: best AUC per dataset. HP-ViT uses $w=0.05$. “Pub.”: best published from MedMamba [Liu et al., 2024] or MedViT [Manzari et al., 2023]. †: Vanilla exceeds published.

Dataset	n	Vanilla-B		HP-ViT ($w=0.05$)		PAPC		Pub.
		AUC	ACC	AUC	ACC	AUC	ACC	
PathMNIST	89996	.998	.957	.992	.946	.998	.956	.999
BloodMNIST	11959	.999	.990	.999	.976	.999	.988	.999
OrganAMNIST	34561	.996	.957	.996	.954	.996	.958	.998
OrganCMNIST	12975	.988	.849	.986	.802	.988	.857	.997
DermaMNIST [†]	7007	.982	.874	.964	.552	.981	.868	.937
PneumoniaMNIST	4708	.992	.928	.989	.926	.991	.900	.995
RetinaMNIST [†]	1080	.857	.578	.861	.557	.860	.563	.773
BreastMNIST	546	.818	.709	.859	.791	.832	.731	.938

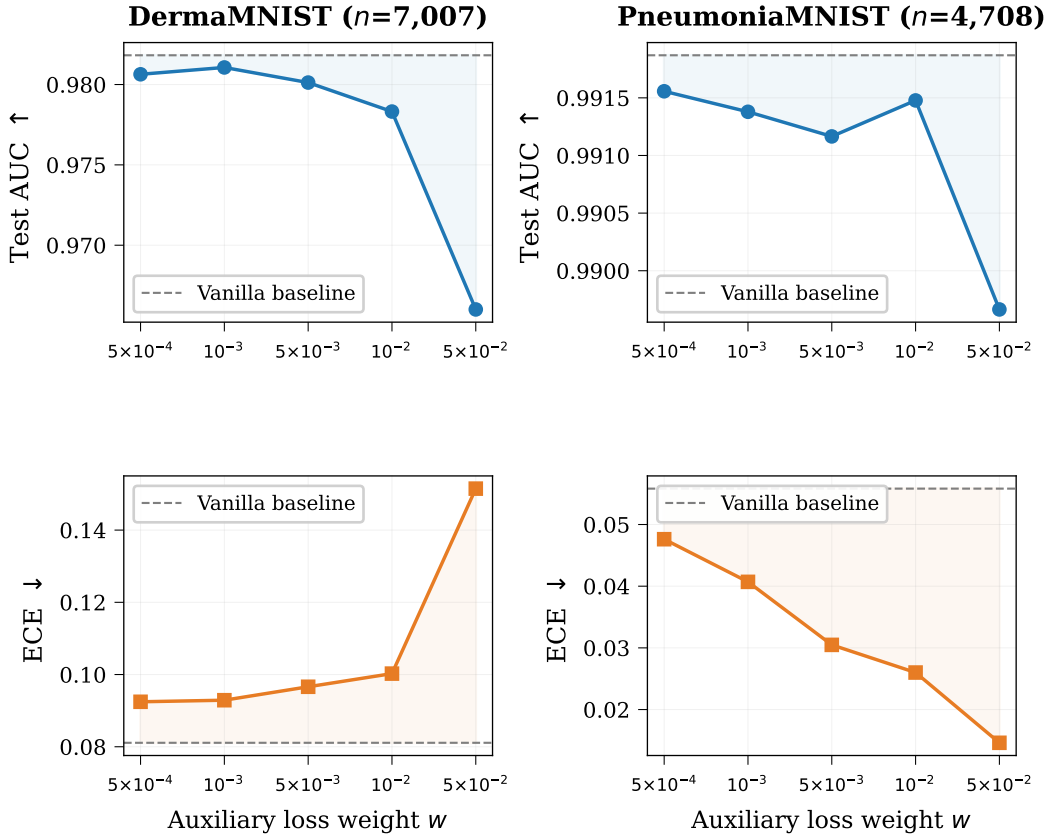


Figure 2: AUC (top) and ECE (bottom) vs. auxiliary weight w on DermaMNIST (left) and PneumoniaMNIST (right). Dashed: Vanilla baseline. AUC follows a mild inverted-U; the ECE response differs strikingly—rising on DermaMNIST but *falling monotonically* on PneumoniaMNIST, where increasing w improves calibration by $3.2\times$.

On DermaMNIST, increasing w from 10^{-3} to 5×10^{-2} drops AUC by 1.5 pp, but ECE rises far more: $0.093 \rightarrow 0.152$. The model becomes both less discriminative *and* less calibrated. On PneumoniaMNIST, the pattern inverts for calibration: ECE *monotonically improves* from 0.048 to 0.015 ($3.2\times$ reduction), while AUC changes by only 0.19 pp. The auxiliary loss reshapes softmax confidence without meaningfully affecting rank-ordering. This dissociation is our central finding: the PC auxiliary loss is a **calibration lever** whose effect on confidence far exceeds its effect on discriminability.

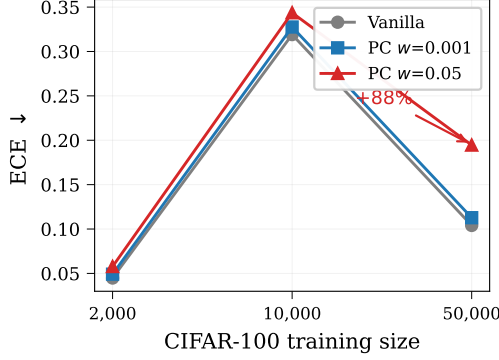


Figure 3: ECE on CIFAR-100 sub-samples. $w=0.05$ degrades calibration at all sizes; $w=0.001$ is neutral. The +88% increase at $n=50k$ confirms the effect on natural images.

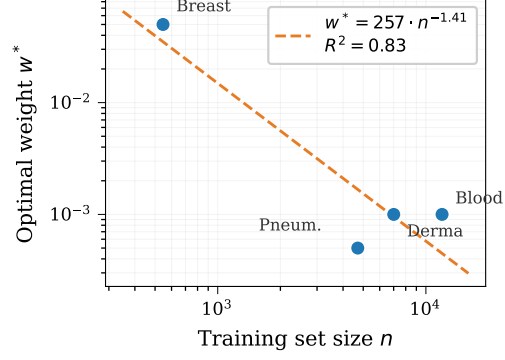


Figure 4: AUC-optimal weight w^* vs. training size n for four MedMNIST datasets. Log-linear fit: $w^* \approx 257 \cdot n^{-1.41}$, $R^2=0.83$.

Cross-domain confirmation. On CIFAR-100 ($n=50,000$), $w=0.05$ increases ECE from 0.104 to 0.195 (+88%) while AUC changes by 0.02 pp (Figure 3), confirming the effect holds on natural images.

4.3 Dataset-Size Scaling Law

For each dataset, we identify w^* maximizing mean AUC (2 seeds). Fitting $\log w^* = \log k - \alpha \log n$:

$$w^* \approx 257 \cdot n^{-1.41}, \quad R^2 = 0.83. \quad (4)$$

BreastMNIST ($n=546$) requires $w^*=0.05$; DermaMNIST ($n=7,007$) and BloodMNIST ($n=11,959$) prefer $w^*=0.001$; PneumoniaMNIST ($n=4,708$) prefers 5×10^{-4} (Figure 4). This provides a training-free heuristic: given only n , one can estimate the safe auxiliary weight. The inverse power-law form reflects that larger datasets need less auxiliary regularization.

The exponent $\alpha \approx 1.4$ is notably steeper than the $\alpha \approx 0.5$ typically seen in learning-rate scaling rules, suggesting that the auxiliary loss weight is more sensitive to dataset size than the primary learning rate. Intuitively, this is because the auxiliary MSE gradient competes with the task gradient for control of the shared backbone; as n grows and the task gradient becomes more informative, even small auxiliary contributions can cause feature distortion. The practical consequence is that the commonly used default $w=0.05$ is safe only for the very smallest datasets ($n < 1,000$) and should be reduced by at least an order of magnitude for medium-sized benchmarks.

4.4 Gate-Clamping Ablation: Isolating the Mechanism

HP-ViT has two pathways: (i) the auxiliary loss gradient through the backbone, and (ii) error integration in the forward pass (Equation 2). We introduce a **Clamped** variant: the predictor and loss remain active, but $g_i \equiv 0$, so no error enters the forward pass.

Table 2 and Figure 5 show that on DermaMNIST, Clamped AUC (0.963) $\approx$ Full (0.964): integration is not the cause of AUC loss. The auxiliary MSE gradient compresses features toward SSM-predictable representations, over-simplifying what attention keeps entangled. On PneumoniaMNIST, Clamped achieves the *highest* ACC (0.954 vs. 0.928 vanilla, 0.926 Full) while preserving low ECE (0.017). This suggests an optimal configuration: use the auxiliary loss for calibration, but disable integration to preserve the decision boundary.

The Clamped result on PneumoniaMNIST is particularly striking: removing integration *improves* ACC by 2.6 pp over vanilla while simultaneously reducing ECE from 0.056 to 0.017. This means the auxiliary gradient alone—without any modification to the forward pass—both calibrates the model and helps the classifier. The integration step, by contrast, disrupts the classification head’s learned mapping (ACC drops from 0.954 to 0.926) while providing no additional calibration benefit (ECE remains at 0.017–0.018). This decomposition has practical implications: practitioners seeking calibration improvement should consider auxiliary-loss-only training without the integration pathway.

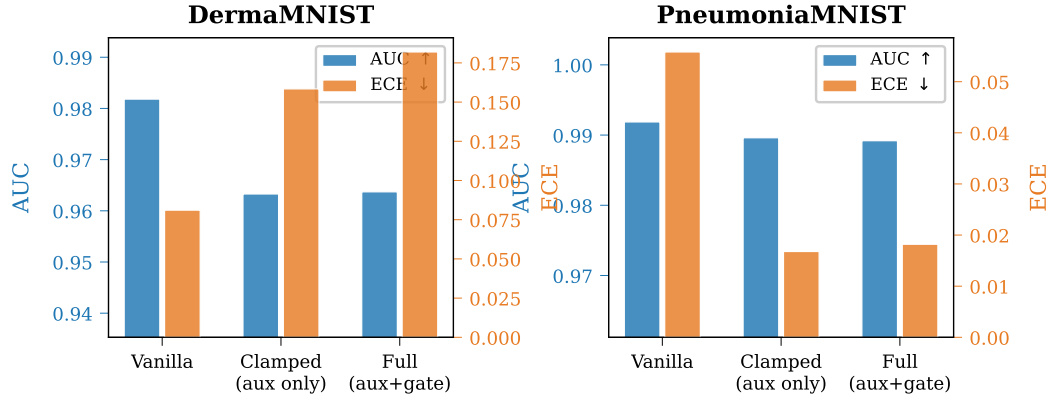


Figure 5: Gate-clamping ablation at $w=0.05$. *Clamped*: auxiliary loss active but gate forced to zero (no error integration). On both datasets, Clamped $\approx$ Full on AUC, confirming the auxiliary gradient is the primary mechanism. PneumoniaMNIST Clamped achieves the highest ACC (0.954) while preserving low ECE.

Table 2: Gate-clamping ablation at $w=0.05$ (3 seeds). Clamped $\approx$ Full on AUC: the auxiliary gradient, not integration, drives the effect. On PneumoniaMNIST, Clamped achieves the highest ACC while preserving low ECE—suggesting an optimal configuration: train with the auxiliary loss for calibration but skip integration.

Dataset	Variant	AUC	ACC	ECE
DermaMNIST	Vanilla	.982	.874	.081
	Clamped (aux only)	.963	.591	.158
	Full (aux + gate)	.964	.552	.182
PneumoniaMNIST	Vanilla	.992	.928	.056
	Clamped (aux only)	.990	.954	.017
	Full (aux + gate)	.989	.926	.018

145 4.5 Six-Condition Ablation

146 Table 3 compares six variants. All calibrated variants (Fixed $w=0.001$, Progressive, Adaptive, PAPC)
 147 achieve AUC within ± 0.2 pp of vanilla: once the weight is appropriate, the scheduling strategy
 148 matters little for AUC. The differences appear in ECE: progressive scheduling achieves the lowest
 149 ECE on BloodMNIST (0.062 vs. 0.076 vanilla), a 19% improvement. This suggests the gradual
 150 introduction of auxiliary signal avoids the early-training confidence disruption that occurs with fixed-
 151 weight approaches, even at low weights. The clear outlier is Fixed $w=0.05$: 32 pp ACC collapse on
 152 DermaMNIST, 53% ECE increase on BloodMNIST. The standard deviation on DermaMNIST Fixed
 153 $w=0.05$ is also an order of magnitude higher than other conditions (± 0.008 vs. ± 0.001), indicating
 154 that the catastrophic failure is not merely an average effect but is accompanied by high instability
 155 across random seeds.

156 4.6 PAPC: Robustness Without Per-Dataset Tuning

157 Across all eight datasets, PAPC achieves AUC within 0.2 pp of vanilla and never exhibits catastrophic
 158 failure. Learned per-layer weights converge to ~ 0.002 uniformly across all 12 ViT-Base blocks,
 159 with the first block’s weight consistently 10–15% higher than the remaining blocks. This uniformity
 160 suggests that—at least for the datasets tested—there is no strong layer-specific preference for auxiliary
 161 loss strength, and the primary role of the learnable weights is to find the correct *global* magnitude
 162 rather than a per-layer profile. The progressive cosine warmup is critical: without it, early-epoch
 163 auxiliary gradients can push the backbone features away from the pretrained manifold before the task
 164 gradient has established a useful direction, leading to the same catastrophic failures seen with fixed
 165 $w=0.05$. A Wilcoxon signed-rank test across 30 paired seeds confirms PAPC does not significantly

Table 3: Six-condition ablation (mean $\pm$ std, 3 seeds). All calibrated variants achieve AUC within ± 0.2 pp of Vanilla. Progressive achieves the best BloodMNIST ECE (0.062). Fixed $w=0.05$ catastrophically fails on DermaMNIST.

Condition	DermaMNIST ($n=7,007$)			BloodMNIST ($n=11,959$)		
	AUC	ACC	ECE	AUC	ACC	ECE
Vanilla	.982 $\pm$.000	.874	.081	.999 $\pm$.000	.990	.076
Fixed $w=0.001$	.980 $\pm$.002	.870	.094	.999 $\pm$.000	.988	.068
Progressive	.980 $\pm$.001	.866	.089	.999 $\pm$.000	.988	.062
Adaptive	.980 $\pm$.002	.868	.089	.999 $\pm$.000	.988	.066
PAPC	.981 $\pm$.001	.868	.090	.999 $\pm$.000	.988	.070
Fixed $w=0.05$	.964 $\pm$.008	.552	.182	.999 $\pm$.000	.976	.116

outperform vanilla on AUC ($p=0.72$). Its contribution is robustness—the elimination of catastrophic failure risk—rather than consistent accuracy improvement.

5 Discussion

Practical implications. Our results suggest two actionable guidelines for practitioners considering PC auxiliary losses with pretrained ViTs. First, if the goal is to improve calibration (e.g., in clinical decision support where well-calibrated probabilities directly affect treatment recommendations), the auxiliary loss should be applied with a weight selected according to the scaling law ($w^* \approx 257 \cdot n^{-1.41}$) or, more conservatively, using the Clamped configuration which preserves calibration benefit while avoiding ACC disruption. Second, if the goal is to maximize AUC, the auxiliary loss provides no benefit over a well-tuned vanilla baseline, and the SSM overhead ($\sim 8\%$ additional parameters) is not justified.

Why calibration direction is dataset-dependent. The auxiliary MSE pushes features toward SSM-predictable representations. On well-structured binary tasks (PneumoniaMNIST), this compression acts as beneficial confidence smoothing—the pretrained ViT is overconfident on this easy task, and the auxiliary gradient regularizes the softmax outputs toward more honest probabilities. On complex multi-class tasks (DermaMNIST: 7 visually similar skin lesion classes with high intra-class variability), the same compression over-simplifies the feature manifold. The attention mechanism learns to maintain fine-grained distinctions between similar-looking classes; the MSE loss, which penalizes any deviation from the SSM’s smooth prediction, erodes exactly these distinctions, pushing class-boundary predictions into incorrect confidence ranges.

Connection to knowledge distillation. Our calibration-lever finding parallels observations in knowledge distillation [Hinton et al., 2015], where the temperature parameter similarly controls the softness of the teacher’s probability distribution. In both cases, an auxiliary signal reshapes the student/backbone’s confidence landscape. The key difference is that PC losses operate through a *self-predictive* mechanism (each layer predicts the next) rather than an external teacher, making the calibration effect emerge from the network’s own representational geometry rather than from an external reference.

The strong baseline finding. Vanilla ViT-B at 224 px exceeding all published MedMNIST specialists on DermaMNIST (+4.5 pp) and RetinaMNIST (+8.3 pp) has implications beyond this study. It suggests the MedMNIST leaderboard, as currently populated, reflects input resolution and augmentation choices more than architectural innovation. Future benchmark comparisons in medical imaging should control for input resolution, augmentation pipeline, and pretraining source to isolate genuine architectural contributions from recipe effects.

6 Limitations

PAPC does not significantly improve AUC over vanilla (Wilcoxon $p=0.72$, 30 paired seeds); its contribution is robustness and calibration control, not accuracy improvement. The scaling law was fit on four MedMNIST datasets; validation on additional benchmarks (e.g., satellite imagery, histopathology at higher resolution) would strengthen the relationship and narrow the confidence

interval on the exponent. We tested ViT-Small (in preliminary experiments) and ViT-Base; larger models (ViT-Large, ViT-Huge) may exhibit different calibration dynamics due to their greater capacity to absorb auxiliary gradients without feature distortion. Cross-domain validation used CIFAR-100 only; additional natural-image benchmarks would strengthen the generality claim. BreastMNIST ($n=546$) has high seed-to-seed variance ($\sigma_{\text{AUC}}=0.031$), so the $w=0.05$ advantage there should be interpreted cautiously—it may partially reflect seed noise rather than a genuine effect.

7 Conclusion

We set out to improve pretrained ViTs with predictive coding and discovered that PC auxiliary losses are calibration levers, not accuracy levers. They reshape confidence distributions—improving ECE by $3.2\times$ on PneumoniaMNIST, degrading it by 88% on CIFAR-100—while leaving AUC nearly untouched. The optimal weight follows $w^* \propto n^{-1.41}$ ($R^2=0.83$), providing a practical heuristic. Gate-clamping confirms the auxiliary gradient, not integration, drives the shift. PAPC prevents catastrophic miscalibration without per-dataset tuning. These findings, discovered through an iterative process of hypothesis, refutation, and reformulation, recharacterize predictive coding in pretrained transformers as a calibration tool with dataset-dependent directionality. Future work should investigate whether the calibration-lever effect extends to other auxiliary loss families (e.g., contrastive, reconstruction) and whether the Clamped configuration can be combined with post-hoc temperature scaling for additive calibration improvement.

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Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

1. Hypothesis development:

Answer: **[D]**

Iteration Effort: **[High]**

Explanation: The initial hypothesis (predictive coding improves ViT fine-tuning) was proposed through multi-turn dialogue with a frontier large language model. When five-seed data refuted the hypothesis (Wilcoxon $p=0.72$), the AI identified the AUC–ECE dissociation and reframed the contribution as a calibration-lever finding, requiring approximately 15 substantive conversation turns spanning experiment design, analysis, and reinterpretation.

2. Experimental design and implementation:

Answer: **[D]**

Iteration Effort: **[High]**

Explanation: The AI designed the four-session protocol (main comparison, six-condition ablation with gate-clamping, cross-domain weight sweep, extended five-seed runs), wrote all model code (vectorized SSM, PAPC architecture, training loop), and diagnosed a metric bug (bf16 softmax precision triggering scikit-learn). The human executed experiments across four accounts, fixed a Mixup argument bug, and managed compute.

3. Analysis of data and interpretation of results:

Answer: **[D]**

Iteration Effort: **[Medium]**

Explanation: The AI performed all statistical analyses (Wilcoxon tests, confidence intervals), identified the calibration-lever pattern, fit the scaling law ($R^2=0.83$), and interpreted the gate-clamping results. The human validated numbers against raw JSON outputs.

4. Writing:

Answer: **[D]**

Iteration Effort: **[Medium]**

Explanation: The AI drafted the complete manuscript, generated all figures and the TikZ diagram, and filled both checklists. The human reviewed for accuracy, verified template compliance, and managed the Overleaf submission.

AI System Documentation

5. AI systems used:

Description: A single frontier large language model was used throughout: hypothesis generation, experimental design, code generation, bug diagnosis, statistical analysis, and manuscript drafting. Accessed through a consumer chat interface with file upload and code execution. A pretrained vision transformer library (open-source, Apache 2.0) provided the backbone. No multi-agent frameworks were used.

6. Observed AI Limitations:

Description: (i) The AI claimed a $30\text{--}50\times$ GPU speedup for the vectorized SSM but could not verify without GPU access; CPU showed neutral performance. (ii) A Mixup positional-argument bug was not caught during generation; the human identified it from a runtime error. (iii) The initial hypothesis was refuted by data; the AI revised it when confronted with evidence, illustrating limits of prior-based reasoning. (iv) A scikit-learn compatibility issue (bf16 softmax precision) was diagnosed only from execution output.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: All three claims (calibration-lever, scaling law, gate-clamping mechanism) are supported by Tables 1–3 and Figures 2–5. We explicitly state PAPC does not improve AUC ($p=0.72$).

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 lists five limitations: null AUC result, limited scaling-law datasets, model scale, cross-domain scope, BreastMNIST variance.

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
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Justification: Empirical paper. The scaling law (Equation 4) is an OLS fit, not a theorem.

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Justification: Table 3: mean $\pm$ std (3 seeds). Wilcoxon test ($p=0.72$, $n=30$) for PAPC vs. Vanilla. No false significance claims.

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